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# Truthful online double auctions for on-demand integrated ride-sourcing platforms

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ABSTRACT

Integrated platforms are an emerging model of business that integrates ride-sourcing services provided by multiple companies on a single platform. It reduces market fragmentation and brings benefits to both travelers and participating independent transport service providers. To ensure the sustainability of such platforms, the main challenge lies in efficient matching between travelers and ride-sourcing companies, so that both sides obtain non-negative utility. In this study, we propose a double auction based framework to solve the matching and pricing problems for on-demand integrated ride-sourcing platforms. To solve the real-time matching problem, we develop two online mechanisms with a critical price dynamic updating algorithm and a greedy algorithm. The proposed mechanisms are guaranteed to achieve the desired economic properties, and their efficiency is evaluated through numerical studies.

## 1. Introduction

With the rapid development of on-demand ride-sourcing companies, competition in the ride-sourcing market has become fierce. Big companies, such as Uber, Lyft and Didi Chuxing, have taken the lead and gained a large market share, while small and medium-sized platforms continually enter the market and struggle to attract users. To avoid a monopoly trend in the ride-sourcing market, a new business model that aggregates small and medium-sized platforms on a "platform of platforms", known as an integrated ride-sourcing service platform (IRSP) has emerged, e.g., AutoNavi and Meituan in China. Those integrated platforms provide travelers with access to transport services from multiple transport service providers (TSPs) through a single platform. For example, AutoNavi integrates the services of T3Go, Shenzhou and other ride-sourcing companies and provides its users with a wide variety of options from different TSPs. The benefits of integrated platforms are multifold. Small TSPs can use the platform to reach a larger customer pool, which has the potential to increase their resource utilization rate and reduce their operational risk. The ride-sourcing market can alleviate market fragmentation, reduce matching friction and further shorten the waiting times of both service vehicles and customers by pooling demand and mobility resources (Zhou et al., 2021). Ride-sourcing customers can access diverse mobility services without switching between platforms, meanwhile enjoying a rapid matching response from the platform.

An IRSP forms a two-sided market, in which TSPs are mobility resource sellers and travelers are mobility service buyers. A platform operator takes the broker role, purchasing resources from TSPs and then selling services to travelers in response to their demand. To ensure the sustainability of such a two-sided platform, the key challenge lies in allocating travelers to TSPs effectively so that travelers' individual needs are satisfied, TSPs gain non-negative profits and system efficiency is achieved. The resource allocation problem has been investigated thoroughly in the literature in various contexts, among which market equilibrium theory is one of the most prevalent approaches. Market equilibrium theory captures the relationship between supply and demand, and has been widely applied to solve resource allocation and pricing problems in transportation engineering, e.g., in the taxi market (e.g. Yang & Wong, 1998, Qian & Ukkusuri, 2017), ridesharing market (e.g. Wang et al., 2018, Ke et al., 2020, Di & Ban, 2019, Ding et al., 2022) and multimodal transport systems (e.g. Huang et al., 2000, Li et al., 2007, Jian et al., 2020). Researchers primarily assume that agents' utility functions are known inputs. However, in practice the planning agency may not hold complete information about agents. Even when the price is simulated by a sequence of changes in response to market supply

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and demand, an agent may consistently report its demand in accordance with the utility function it wants to convey to the market, instead of its true utility function. Misreporting preferences may help agents to increase their own utility but can severely harm the efficiency of the system.

For the demand side, to characterize travelers' personalized travel needs and valuations, researchers conventionally use operational data or stated preference surveys. However, operational data are too limited to draw representative conclusions for such a new business scheme. Stated preference surveys are powerful tools for understanding individuals' preferences for innovative services, but they require substantial investment of time and resources and cannot capture travelers' heterogenous travel requirements in a timely manner. For the supply side, TSPs' operational parameters (i.e., TSPs' operational costs) are also not readily available if they do not cooperate with the IRSP operator. TSPs are selfish entities that may strategically misreport operational costs to obtain better allocations, resulting in a fairness issue and efficiency loss. Therefore, it is necessary to design a resource allocation mechanism that takes strategic travelers and TSPs into account.

In addition, due to the provision of on-demand services to travelers, the IRSP operates within a dynamic environment. This entails that travelers' requests and TSPs' idle vehicles arrive at the platform sequentially in an online manner. The IRSP operator must promptly make a decision regarding matching and pricing upon receiving a request, without possessing knowledge of future supply and demand conditions. Therefore, there is a need for an online mechanism that can dynamically solve the matching and pricing problem, ensuring both efficiency and desired economic features.

In order to tackle the aforementioned challenges, we develop an online framework that utilizes a double auction mechanism for matching and pricing within an integrated on-demand ride-sourcing platform. First, on the demand side, this framework takes into account the strategic behavior of travelers. The utilization of auction-based mechanism design offers a means to elicit travelers' truthful valuations and personalized travel requirements, thereby addressing the problem of incomplete information in the allocation of resources (Myerson & Satterthwaite, 1983). Second, on the supply side, the auction serves two purposes: accurately capturing the truthful operational expenses of TSPs and safeguarding their interests through individual rationality for TSPs. During the initial phases of their establishment, IRSPs, such as AutoNavi, employed a strategy of subsidizing participating TSPs and enticing travelers with competitive pricing in order to actively acquire a larger market share and facilitate the growth of their businesses. Nevertheless, as the development reaches a state of stability, AutoNavi not only ceases to subsidize TSPs but also urges them to uphold competitive pricing in order to continuously attract consumers. Due to the inherent power imbalance between TSPs and the IRSP, TSPs frequently find themselves compelled to conform to the price policies set by IRSPs, even if it results in financial losses for the TSPs. This issue is widely observed inside the platform economy. In order to remain competitive, IRSPs endeavor to prioritize the satisfaction of buyers' requirements, sometimes at the expense of sellers' interests (Qu and Deng, 2022). Unreasonably low payments to TSPs in light of a significant increase in the price of fuel will lead to a strike by TSPs' drivers (Ikpoto, 2023), which is detrimental to the IRSP's sustainable development as drivers leave the platform and TSPs suspend cooperation to avoid deficits, ultimately harming the welfare of travelers. By employing an auction mechanism, we let TSPs bid their operational costs truthfully and ensure that they are guaranteed a non-negative utility. This protects the interests of TSPs and lays a solid foundation for cooperation by preventing the IRSP from arbitrarily reducing the price without justification.

Due to its straightforwardness and efficacy, auction has been broadly implemented in various practical applications, including Facebook's advertising system and vehicle registration mark auction. To respond to travelers and TSPs in real time, we develop two online mechanisms that are capable of achieving a commendable level of efficiency. The primary contributions of this study are summarized as follows:

- We develop a double auction framework to elicit travelers' truthful valuations and requirements and TSPs' actual operational costs, thereby establishing the basis for maximizing social welfare. The proposed mechanisms can be applied to real-world two-sided IRSP matching and pricing problems involving multi-minded travelers and TSPs with multi-unit capacity.
- To ensure the viability of the proposed framework, we develop two online mechanisms based on a critical price updating algorithm and a greedy algorithm that provide real-time responses to travelers and TSPs. Both mechanisms achieve the desired economic properties such as incentive compatibility (IC), individual rationality (IR) and weakly budget balance (WBB). We compare their performance with the first-arrive-first-serve matching and another online mechanism based on padding method in numerical studies.
- Extensive numerical experiments are conducted to demonstrate the benefits of introducing IRSP and to assess the performance of the proposed online mechanisms. Results indicate that both critical price dynamic updating and greedy algorithms achieve higher approximation ratios on social welfare than first-arrive-first-serve matching. It is demonstrated that both online mechanisms run in polynomial time, indicating their applicability.

The rest of this paper is organized as follows. Section 2 reviews related work. The double auction model and online environment setting are discussed in Section 3. Section 4 presents two mechanisms to solve the online double auction problem (the third one is presented in the supplementary material). Numerical studies are conducted in Section 5, followed by the conclusions summarized in Section 6.

## 2. Literature review

In this section, we briefly review work related to our research context – integrated ride-sourcing platform, research question – two-sided market matching problem in transportation, methodology – double auction mechanism design and summarize the research gap.

### 2.1. Integrated ride-sourcing platform

The benefits promised by IRSP has promoted its practical implementation. However, studies of such an integrated platform are very few. Previous studies have mainly focused on quantifying the economic and operational benefits of IRSPs and discussed the optimal operational strategies of individual TSPs within different market structures. Zha et al. (2016) studied the monopoly and duopoly ride-sourcing markets and found that the prices at Nash equilibrium are not necessarily lowered by competition and that social welfare may suffer a decline in the duopoly market because of the less efficient matching technology. They further suggested that the regulator might rather encourage the merging of the ride-sourcing companies. Séjourné et al. (2018) investigated quantitatively the efficiency loss when demand is exogenously split between multiple platforms. The review paper by Wang and Yang (2019) discussed the findings regarding competition between multiple ride-sourcing companies. These studies demonstrated that under certain market scenarios, the integration of multiple ride-sourcing companies can bring economic benefits to ride-sourcing companies.

Regarding the operational strategies of individual TSPs given the existence of an IRSP, Cohen and Zhang (2022) found that if competing ride-sourcing platforms engage in a profit-sharing contract by introducing a new joint service, a well-designed profit-sharing contract will benefit both travelers and platforms especially when demand exceeds supply. Zhou et al. (2021) studied the impact of integrating ride-sourcing platforms in a competitive market and concluded that platform-integration can increase the total realized demand and social welfare at Nash equilibrium and social optimum, but may not

necessarily guarantee a higher profit for individual ride-sourcing companies when the vehicle supply is sufficiently large, and/or the market is considerably fragmented. These studies indicated that an IRSP may not benefit all TSPs, which highlights the importance of developing an operating mechanism that can ensure non-negative profit to participating TSPs.

### 2.2. Two-sided market supply-demand matching in transportation

There is a substantial body of research on two-sided market supply-demand matching problems, most of which focus on matching travelers and drivers with single or multiple competitive TSPs (Ke et al., 2019; Lyu et al., 2019; Zha et al., 2018). These studies have set various objectives for the matching problem, e.g., minimizing travelers' total waiting time, minimizing the operational costs or maximizing the profit of TSPs. The rapid development of ride-sourcing companies has boosted studies of the multi-homing behavior of both travelers and drivers. Jeitschko and Tremblay (2020) extracted the equilibria with multi-homing and single-homing on either side of the market. They found that when the decision on whether to go multi-homing is made endogenously, multi-homing induces lower prices, attracting more agents. Bernstein et al. (2021) explored the multi-homing behavior of drivers and determined that the complete multi-homing of all drivers harms everyone's utility although individual drivers may have an incentive to choosing multi-homing. Based on this finding, the authors concluded that multi-homing should be discouraged. Ma et al. (2022) designed a spatio-temporal pricing mechanism to match drivers and riders to trips for the ridesharing platform, using dynamic prices to achieve a supply-demand equilibrium. Guo et al. (2023) investigated the dispatching strategies for online ridesharing platforms from a complete sustainability perspective.

Unlike the above research, where the ride-sourcing market consists of travelers, drivers and TSPs, in our model, we assume drivers subscribe to a single TSP, so that the matching problem faced by the IRSP is concerned only with travelers and TSPs. This is common in reality that ride-sourcing drivers (such as Uber and Didi Chuxing drivers) only serve one TSP and follow the matching decisions made by the TSP. Also, previous two-sided market studies mainly discussed the outcome at Nash equilibrium rather than the matching strategies between supply and demand at the operational level. The papers by Schwarzstein and Schouery (2023) and Ding et al. (2023) applied auction-based mechanisms to solve matching for ridesharing and ride-sourcing platforms, respectively, which are the closest ones to our problem. However, their mechanisms are one-sided, where only travelers are allowed to bid for the service, neglecting the strategic behavior of TSPs.

### 2.3. Double auction mechanism design

A double auction-based mechanism is the main methodology we rely on to solve the matching problem between travelers and TSPs. Double auction has wide applications in the transport context, e.g., the transport procurement problem in the logistics market (Xu & Huang, 2013; Xu & Huang, 2014) and the parking space matching problem (Xiao et al., 2018; Shao et al., 2020).

Early double auction studies usually made two assumptions: single-minded bidders and single output restriction (e.g., Babaioff & Walsh, 2005; Chu & Shen, 2006, 2008). The single-minded assumption requires bidders to be only interested in a single specified bundle of items. Bidders will obtain a certain value if they win the whole bundle, otherwise they obtain zero value. The single output restriction requires that each seller can only submit bids on a single unit of one commodity. However, our problem violates both assumptions. In an IRSP, travelers have multi-dimensional requirements and thus associate different values with different types of service, meanwhile TSPs own multiple service vehicles on multiple routes. If the single output restriction is simply relaxed to the multi-unit setting without any modification to the double auction mechanism, the budget balance property will not hold. Since the single output restriction is not realistic in practice, many studies have discussed the design of a multi-unit double auction. Chu (2009) later extended the single-minded multi-unit double auction to a market where buyers demand some bundles with various values and suppliers have multiple units of one commodity.

Of these double auction studies, the trade reduction approach (Babaioff & Walsh, 2005) and multi-stage mechanism (Chu & Shen, 2008; Chu, 2009) are two classic approaches. Huang and Xu (2013) designed multi-unit trade reduction mechanisms for a bilateral exchange transportation marketplace with single-minded shippers (buyers) and carriers (sellers) with multi-unit items and found that higher utilities are realized in buyer and seller augmented mechanisms. Xiao et al. (2018) developed novel truthful double auction mechanisms based on the padding method for shared parking management where demanders are single-minded and suppliers have multi parking resource units. The mechanism they designed guarantees IC, IR, BB and achieves asymptotic efficiency, but does not apply to the setting of multi-minded buyers and sellers with multi-unit items.

In terms of online problems, a number of studies have applied the online double auction to a wide range of contexts. For wireless communications, Wang et al. (2010) made the first attempt to formulate a general framework of a truthful online double auction for spectrum allocation. Taking into consideration that spectra are heterogeneous, Feng et al. (2012) proposed a truthful, individual rational and budget-balanced double auction mechanism that allows buyers to express their personalized preferences explicitly. In the energy trading market, An et al. (2017) designed a strategy-proof online double auction scheme to balance the energy among microgrids' electric grid systems. Yassine et al. (2019) proposed double auction models to incentivize autonomous electric vehicle agents to participate in the market based on their time preferences and energy pricing. In the cloud computing market, Dibaj et al. (2020) developed a dynamic online double auction mechanism capable of modeling cloud ecosystems realistically and coined the term sustainable pricing for cloud computing services. Patel et al. (2021) proposed an online truthful double auction technique for balancing the multi-objective trade-offs between energy, revenue, and performance in Infrastructure-as-a-Service (IaaS) clouds. However, research into online double auction design with relaxation on single-minded bidders and single output restriction assumptions is lacking.

### 2.4. Research gaps

The majority of current studies on transport platform integration in a two-sided market with competing ride-sourcing companies have compared the Nash equilibrium with and without an integrated platform and discussed the potential economic benefits of such an innovative business mode. Theoretical studies of matching strategies between travelers and TSPs for IRSPs at the operational level are still limited. To guarantee the efficiency and sustainability, complete information about travelers' heterogeneous demand functions and the true valuations of both travelers and TSPs are required while designing matching rules. Double auction mechanisms promise to reveal such information. However, unlike reviewed studies that applied a double auction to solve the two-sided matching problems in the field of transport, our problem is more complex in that travelers are multi-minded travelers with heterogeneous valuations of different service types and TSPs have multi-unit capacity. Few studies have investigated the double auction mechanism design for multi-minded buyers and sellers with multi-unit resources. Moreover, considering that the IRSP operates in a dynamic online environment, classic double auction mechanisms are not applicable directly to the online setting due to the long computation time and uncertainty of supply and demand. Online double auction mechanism for IRSPs that can elicit truthful information of travelers and TSPs and make real-time responses has not been investigated yet. In the following

sections, we will present the model and solution algorithms developed to solve this problem.

## 3. Problem statement

We consider an IRSP operating in a transport network with multiple OD pairs in an online environment. Several TSPs sell heterogeneous mobility services to the platform at their declared operational costs and travelers buy services from the platform. The IRSP operator needs to match travelers to TSPs such that travelers' travel requirements are satisfied, meanwhile TSPs can earn non-negative profits through providing mobility services. As TSPs and travelers arrive at the platform dynamically, the matching should be conducted by the IRSP operator promptly without knowing the future supply and demand. To solve this matching problem, we need to obtain travelers' travel requirements and valuations as well as TSPs' operational costs as the model inputs. In this section, we present the online double auction model.

### 3.1. Description and notation

Let $I$ denote the set of travelers and $J$ denote the set of TSPs. In the double auction, TSPs and travelers are regarded as sellers and buyers, respectively. They submit their bids to the IRSP, which is also known as the auctioneer. The relationship of agents and the auction process are presented in Fig. 1.

Each traveler only submits one bid while each TSP can submit multiple bids for each idle vehicle. We assume each TSP can operate several types of transport services $s \in S_{O,D}$ for each OD pair $\{O,D\} \in H$, where $S_{O,D}$ denotes the set of service types between OD pair $\{O,D\}$ and $H$ denotes the set of OD pairs[1] in the transport network. Different service types vary in rider capacity and comfort level. This assumption reflects the real-world operation of ride-sourcing companies, e.g., Uber provides

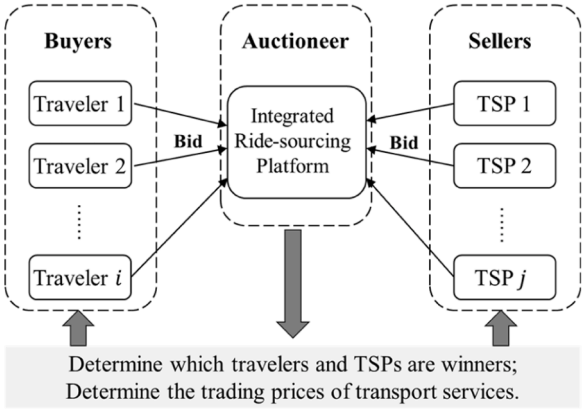



**Fig. 1.** Illustration of transport service market.

UberX for affordable daily rides and UberBlack for premium rides with luxury cars. We use travel time $t_s$[2] and rider capacity $r_s$ to characterize different service types.

Let $K$ represent the set of bids from all TSPs and $K^j$ represent the set of bids from TSP $j$. A bid $k \in K$ includes the following information: submitted by TSP $j_k$, origin $O_k$, destination $D_k$, travel time $t_k$, rider capacity $r_k$, and declared operational cost per vehicle $C_k$. Rider capacity $r_k$ is the maximum allowable number of shared riders in a vehicle. Bid $k$ is submitted by TSP $j_k$, then we have $k \in K^{j_k}$. The information of bid $k$ is denoted by $B_k = \{j_k, O_k, D_k, t_k, r_k, C_k\}$. Note that for a certain service type of a certain OD pair, the travel time and rider capacity are fixed, no matter which TSP operates it. We have $t_k = \sum_{s \in S_{O_k,D_k}} \delta_k^s t_s$, $r_k = \sum_{s \in S_{O_k,D_k}} \delta_k^s r_s$, $\forall k \in K$, where $\delta_k^s = 1$ if bid $k$ is with service type $s$ of OD pair $\{O_k, D_k\}$, otherwise $\delta_k^s = 0$. We assume that throughout one operating time span, a TSP is allowed to bid only one operational cost for vehicles with same service type and same OD pair. This is consistent with reality because the operational cost should be stable over a short time span, e.g., 20 min. TSPs can submit different operational costs for multiple time spans, but in each time span, the operational cost should be the same and we consider the resource allocation problem for each time span separately. Nevertheless, the operational cost is not guaranteed to be credible and it equals the real one only when TSP bids truthfully.

Each bid $B_i$ from traveler $i \in I$ consists of the following information: trip origin $O_i$, destination $D_i$, bidding price $b_i$, preferred travel time $T_i$ and preferred number of shared riders $R_i$. In reality, traveler's requested travel time is usually not a hard constraint, in other words, some travelers can tolerate a certain delay in arrival time. However, violating traveler's preferred travel time will lead to a loss in the value they can obtain, known as late-arrival penalty. Such penalty on late arrival varies across different travelers. Some travelers are very sensitive to time and have high penalty, while some have higher tolerance on late arrival, leading to a low penalty. We use $\gamma_i$ to denote the unit monetary penalty on late-arrival of traveler $i$. The higher the value is, the more sensitive the traveler is to on-time arrival. Another personalized requirement is the number of riders they are willing to share the trip with, denoted as $R_i$. Similar as the preferred travel time, the preferred number of shared riders is a soft constraint and the violation on this requirement leads to a penalty with $\beta_i$ monetary unit. To this end, a bid from traveler $i$ includes the following personalized requirements: $B_i = \{O_i, D_i, b_i, T_i, R_i, \gamma_i, \beta_i\}$[3]. Travelers values different service types variously. Let $V_{is}$ denote the real value of traveler $i$ obtained from traveling with service type $s$, which can be calculated as:

$$V_{is} = V_i - \gamma_i \max\{0, t_s - T_i\} - \beta_i \max\{0, r_s - R_i\} \forall i \in I, \ \forall s \in S_{O_i,D_i} \quad (1)$$

where $V_i$ is the true willingness-to-pay of traveler $i$. Both travelers' true willingness-to-pay and TSPs' true operational costs are private

[1] Note that the ride-sourcing trips can have their origin and destination at any location inside the city, resulting in a vast number of OD pairs throughout the transportation network. Under this context, our proposed online mechanisms can still address the matching and pricing problems for integrated platforms. One classic approach is to divide the city into smaller zones at first and transform the actual OD pairs into zone-based OD pairs, instead of implementing the online mechanism directly. Such simplification drastically reduces the number of OD pairs, and the online mechanisms can be implemented either at the city level by utilizing zone-based OD pairs or at the zone level by employing actual OD pairs. Partitioning the city into grids, such as hexagons, has been widely applied in research and practice, for example, supply-demand forecasting of ride-sourcing services (Ke et al., 2018) and Uber's H3 geospatial indexing system.

[2] The travel time for each service type is assumed to be a general expectation of the entire trip, and it can be estimated from historical operational data and travel information Apps, such as Google Maps. Since the length of each time span is considerably short, e.g., 20 minutes, one can estimate the average travel time using Google Maps and this estimation is assumed to be accurate in our study.

[3] Note that travelers may not know their $\gamma_s$ and $\beta_s$ precisely. To characterize these parameters, the platform operator can ask them several questions, e.g., how much would their willingness-to-pay (WTP) decrease if they are late for five minutes. Then $\gamma_s$ equals the decrease in WTP divided by five minutes. Another way is to learn travelers' preferences from historical bids to calibrate these parameters. For example, $\gamma_s$ could be the difference in WTP divided by the difference in requested travel time between the highest and the lowest bids. Zhang et al. (2015) categorized bidders into three typical valuation types, each with a corresponding valuation function where the parameters are calculated from historical data. Initially, the platform may obtain inaccurate penalty parameters of travelers, but as they bid more often, this private information will become more accurate.

information, which equal the bidding prices $b_i$/costs $C_k$ only when travelers and TSPs bid truthfully.

In the online environment, bids from both travelers and TSPs arrive to the platform sequentially. The IRSP lacks knowledge regarding the specific timing of agents' bids beforehand. Upon receiving a bid from an agent, the platform must promptly determine whether to accept the bid and the corresponding trading price. It is assumed that travelers and TSPs will not cheat on bidding time. TSPs possess the ability to predict the estimated completion time of a trip with an occupied vehicle, enabling them to submit bids in advance. This assumption of "no cheating on bidding time" is plausible. For travelers, late bidding causes their arrival time to be delayed, which negatively impacts their utilities. Untimely bidding by TSPs results in idle vehicles, which wastes mobility resources. Despite the potential negative consequences, certain agents may persist in engaging in intentional late bidding in order to enhance their overall benefits. However, it is widely acknowledged that non-myopic agents constitute a minority in general. To ensure the designed mechanism can match the majority efficiently and reliably, this assumption is made in our study. Similar assumption was made in previous online double auction designs, e.g., Dibaj et al. (2020) stated that agents report their arrival and departure times to the auctioneer with the hope of completing their trade within this time window, indicating there is no bidding time cheating. We will consider the relaxation of this assumption in our future work. Besides, travelers and TSPs are not allowed to cancel their bids, i.e., they cannot leave the system on their own accord or refuse the IRSP's matching decision and rebid in the same operating time span. In previous online double auction studies, the assumption that rebidding is prohibited has been widely accepted, e.g., Wang et al. (2010). In practice, some TSPs, including Uber and Didi Chuxing, impose additional penalty fees to deter travelers from canceling bids on purpose. Once being rejected by the IRSP, travelers can select alternative travel options other than the IRSP, such as taxi. TSPs may also choose to serve their own customers after being rejected, rather than wait for orders assigned from the IRSP.

### 3.2. Model formulation

Our goal is to maximize the social welfare through optimizing the matching decision for travelers (denoted as $x_{is}$ for traveler $i$ if matched to service type $s$) and for TSPs (denoted as $y_k$ for bid $k$ is accepted). The social welfare is defined as the sum of utilities of all agents, including travelers, TSPs and the IRSP in the auction. Traveler's utility equals his or her real value obtained from using the service minus the payment. TSP's utility equals the revenues received from the platform for providing services minus the operational costs. The platform's utility is the difference between the total payments collected from travelers and the total expenditure distributed to TSPs. Adding up these three groups' utilities, the pricing terms can be cancelled out, such that the social welfare equals the real values of all accepted travelers less the operational costs of all winning TSPs. If all agents bid truthfully and all the bids in the operating time span are known to the IRSP, the optimal matching can be solved from the following social welfare maximization problem, denoted as $P(I,K)$:

$$)\overline{P(I,K)} \quad Z(I,K) = \max_{x_{is},y_k} \sum_{i\in I} \sum_{s\in S_{O_i,D_i}} V_{is}x_{is} - \sum_{k\in K} C_k y_k \tag{2}$$

*Subject to*

$$\sum_{s\in S_{O_i,D_i}} x_{is} \le 1 \forall i \in I \tag{3}$$

$$\sum_{i\in I} x_{is} \le \sum_{k\in K} r_k y_k \delta_k^s \forall s \in S_{O,D}, \{O,D\} \in H \tag{4}$$

$$x_{is} \in \{0,1\} \forall i \in I, s \in S_{O_i,D_i} \tag{5}$$

$$y_k \in \{0,1\} \forall k \in K \tag{6}$$

where the binary variable $x_{is} = 1$ if traveler $i$ is matched to service type $s$ and the binary variable $y_k = 1$ if bid $k$ is accepted by the IRSP. Constraint (3) ensures that each traveler receives at most one type of transport service. Constraint (4) is the capacity constraint, requiring that for a certain service type of a certain OD pair, the number of accepted travelers does not exceed the maximum number of riders of vehicles ordered from TSPs. Constraints (5) and (6) set the feasible domains of decision variables $x_{is}$ and $y_k$.

This linear programming is an offline (static) problem since all the bidding information is given as input. However, in practice, bids from travelers and TSPs arrive sequentially at the IRSP, and the platform operator must make immediate and irrevocable decisions regarding whether to accept travelers and TSPs without the knowledge of future supply and demand. Here, we regard the optimal solution to $P(I,K)$ as a benchmark as it is efficient, but impossible to be realized in the online case. We will introduce two online mechanisms that strive to maximize the objective function of $P(I,K)$ in Section 4. Next, we discuss the pricing schemes. To accomplish this, three important economic properties must be guaranteed: (1) incentive compatibility, which ensures bidding truthfully is the dominant strategy of all agents; (2) individual rationality, which ensures all winning agents obtain non-negative utility from the auction; and (3) weakly budget balance, which ensures that the total payments collected from travelers exceed or equal the total expenditures distributed to TSPs, so that the IRSP does not subsidize the auction. Unfortunately, even for the offline problem, it has been proved that these three properties cannot hold with social welfare maximization simultaneously (Myerson & Satterthwaite, 1983). Hence, the efficiency will suffer a loss to guarantee three economic properties.

For the offline problem, Chu (2009) introduced a truthful, individually rational and weakly budget-balanced mechanism that achieves asymptotic social welfare maximization based on the padding method. This mechanism is a classic and effective method for solving resource allocation, and it has been adopted in numerous studies (Chen et al., 2016; Cheng et al., 2016; Xiao et al., 2018). The main idea is to introduce a set of virtual padding travelers with extremely high bidding prices. These virtual padding travelers intensify the competition between travelers, which guarantees the incentive compatibility of travelers. In the meantime, these virtual padding travelers help to reduce the budget deficit in the multi-unit double auction by influencing the trading prices. Chu (2009) demonstrated that the mechanism is asymptotically efficient given sufficient numbers of buyers and sellers. However, this mechanism is not applicable to our online problem for two reasons. Firstly, to perform matching and pricing, it is necessary to have complete information of all travelers' and TSPs' bids. But as we have claimed before, the IRSP cannot obtain all the bidding information at the beginning of the operating time span. Secondly, the pricing scheme of Chu's method is developed from the VCG-like payment rule, which requires long computation time. VCG (Vickrey, 1961; Clarke, 1971; Groves, 1973) auction is a classic auction mechanism that guarantees individual rationality and incentive compatibility. However, lengthy computation times make it impossible to respond to travelers and TSPs in real time.

## 4. Online double auction mechanisms

In this section, we propose two online mechanisms: a critical price dynamic updating algorithm and a greedy algorithm to solve the online matching and pricing problem, both of which satisfy IR, IC and WBB. For comparison purpose, we also propose an online mechanism with the padding method, which is presented in the supplementary material.

### 4.1. Online mechanism designed with a dynamic updating algorithm

The main idea of the critical price dynamic updating algorithm is to collect the bids in a certain time slot and solve the social welfare maximization problem with these bids. On obtaining the optimal solution, we identify the critical price of the transport service of each service type of each OD pair. Then we regard these critical prices as the threshold prices to determine the acceptation criteria and trading prices for future bids. We then dynamically update the critical prices to capture the change in supply and demand over time. Fig. 2 demonstrates the procedure of this online mechanism, followed by the details in each step.

- Step 1: **Initialization**. Let $\mathbb{T}$ denote the total time span, which is divided to $\mathscr{M}$ time slots, denoted as $\mathbb{t}_1, \mathbb{t}_2, \ldots, \mathbb{t}_m$. The length of each time slot is identical. Initialize the capacity vector $\boldsymbol{W}$ to be zero where each element represents the vehicle capacity for each service type of each OD pair.
- Step 2: **Update critical price**. Collect bids from all agents that arrive in $\mathbb{t}_1$. The sets of travelers and TSPs arrive in $\mathbb{t}_1$ are denoted as $I_{\mathbb{t}_1}$ and $J_{\mathbb{t}_1}$ respectively. The set of bids from TSPs is represented by $K_{\mathbb{t}_1}$. Solve $P(I_{\mathbb{t}_1}, K_{\mathbb{t}_1})$ defined in Section 3 with input of traveler set $I_{\mathbb{t}_1}$ and bid set $K_{\mathbb{t}_1}$ and obtain the critical price vector $\boldsymbol{q}_{\mathbb{t}_1}$. Each element $q_{\mathbb{t}_1}^{O,D,s}$ represents the critical price of OD pair $\{O,D\}$ and service type $s$, which equals the lowest real value of this service type and OD pair among all the winning travelers in the solution to $P(I_{\mathbb{t}_1}, K_{\mathbb{t}_1})$.
- Step 3: **Determine potentially winning bids from TSPs**. Since we allow TSPs to submit their bids in advance, at the beginning of $\mathbb{t}_2$ we can obtain all the bids from TSPs that will arrive in $\mathbb{t}_2$. If the operational cost of bid $k$ is less than or equal to the operational cost of the highest winning bid $k_{\mathbb{t}_1}$ with the same service type and OD pair in the solution to $P(I_{\mathbb{t}_1}, K_{\mathbb{t}_1})$, denoted by $C_{\mathbb{t}_1}^{O,D,s}$, then bid $k$ has the potential to win in $\mathbb{t}_2$. We sum up the capacities of these potentially winning bids to obtain $\overline{\boldsymbol{W}}_{\mathbb{t}_2}$, which represents the upper bound of available capacity in $\mathbb{t}_2$.
- Step 4: **Calculate trading prices**. For a certain service type of a certain OD pair, the trading price $p_{\mathbb{t}_2,J}^{O,D,s}$ for TSPs and $p_{\mathbb{t}_2,I}^{O,D,s}$ for travelers are defined as follows. $p_{\mathbb{t}_2,J}^{O,D,s}$ is equal to the highest operational cost $C_{\mathbb{t}_1}^{O,D,s}$ or the VCG price for this highest bid $k_{\mathbb{t}_1}$, i.e., $p_{\mathbb{t}_2,J}^{O,D,s} \in \{C_{\mathbb{t}_1}^{O,D,s}, C_{\mathbb{t}_1}^{O,D,s} + [Z(I_{\mathbb{t}_1}, K_{\mathbb{t}_1}) - Z(I_{\mathbb{t}_1}, K_{\mathbb{t}_1}\{k_{\mathbb{t}_1}\})]\}$. Step 6 will explain which value to use for different bids. For travelers, the trading price (payment) $p_{\mathbb{t}_2,I}^{O,D,s}$ is equal to the greater one between the critical price $q_{\mathbb{t}_1}^{O,D,s}$ and the unit trading price of the TSP, i.e. $p_{\mathbb{t}_2,I}^{O,D,s} = \max\left\{q_{\mathbb{t}_1}^{O,D,s}, \frac{p_{\mathbb{t}_2,J}^{O,D,s}}{r_s}\right\}$.
- Step 5: **Determine winning travelers**. We handle each traveler's allocation individually in accordance with their arrival order in $\mathbb{t}_2$. First, we calculate the maximal achievable real value for traveler $i$ among the available service types based on $\overline{\boldsymbol{W}}_{\mathbb{t}_2}$, denoted by $V_{is^*}$. Then, traveler $i$ wins the bid if his or her maximal achievable value is greater than or equal to the corresponding trading price, i.e., $V_{is^*} \geq p_{\mathbb{t}_2,I}^{O,D,s^*}$. Accepted travelers will be served in $\mathbb{t}_2$ and charged $p_{\mathbb{t}_2,I}^{O,D,s^*}$ for the service type $s^*$, whereas rejected travelers are removed from the auction and cannot rebid. We update $\overline{\boldsymbol{W}}_{\mathbb{t}_2}$ whenever we accept a new traveler.
- Step 6: **Determine winning bids from TSPs**. The acceptance of TSPs' bids is contingent upon traveler demand. In Step 3, we identified potential winning bids. Then, once a traveler is accepted in Step 5, we must order vehicles of the requested service type based on these potential winning bids, with the vehicles with the lowest operational costs coming first. If the operational cost is less than $C_{\mathbb{t}_1}^{O,D,s}$ for an accepted bid $k$, the TSP receives $C_{\mathbb{t}_1}^{O,D,s}$ for the bid; otherwise, the TSP receives the VCG price $C_{\mathbb{t}_1}^{O,D,s} + [Z(I_{\mathbb{t}_1}, K_{\mathbb{t}_1}) - Z(I_{\mathbb{t}_1}, K_{\mathbb{t}_1}\{k_{\mathbb{t}_1}\})]$ for the bid. Rejected bids are removed from the auction, and TSPs are prohibited from rebidding on rejected vehicles. At the end of $\mathbb{t}_2$, we update $\boldsymbol{W}$ to determine whether there is any remaining capacity. $\boldsymbol{W}$ is passed to $\mathbb{t}_3$, and the capacity of potential winning bids is added to $\boldsymbol{W}$ to obtain $\overline{\boldsymbol{W}}_{\mathbb{t}_3}$.
- Step 7: **Iteration**. Repeat Steps 2–6 for future time slots. Specifically, for all bids arrive in $\mathbb{t}_m$, first solve $P(I_{\mathbb{t}_{m-1}}, K_{\mathbb{t}_{m-1}})$ defined in Section 3 with input of traveler set $I_{\mathbb{t}_{m-1}}$ and bid set $K_{\mathbb{t}_{m-1}}$ to update $\boldsymbol{q}_{\mathbb{t}_{m-1}}$ following Step 2. Trading prices calculation and winner determination can refer to Steps 3–6 based on critical price $\boldsymbol{q}_{\mathbb{t}_{m-1}}$ and capacity vectors $\overline{\boldsymbol{W}}_{\mathbb{t}_m}$. $\boldsymbol{W}$ is updated at the end of each time slot and passed along to the next time slot. The repetition process is conducted till the last time slot.

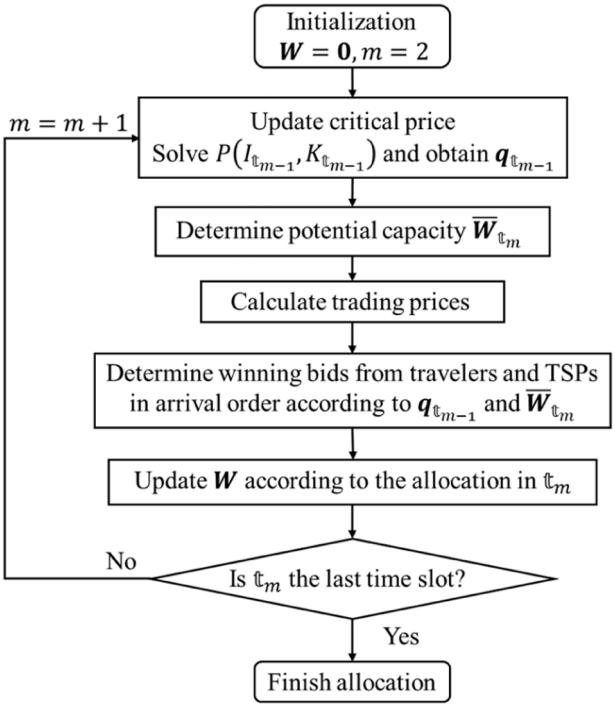



**Fig. 2.** Online mechanism with a critical price dynamic updating algorithm.

The above online mechanism achieves real-time responses to travelers. It uses the critical prices solved from the previous time slot to determine the allocation and pricing in current time slot. Note that winner determination starts from $\mathbb{t}_2$ and the bids arrive in $\mathbb{t}_1$, i.e. $I_{\mathbb{t}_1}$ and $J_{\mathbb{t}_1}$, are all rejected. Since $\mathbb{t}_1$ is substantially small compared to $\mathbb{T}$, the loss in social welfare can be ignored. Let M1 denote the online double auction mechanism designed with the critical price dynamic updating algorithm. We claim that M1 achieves individual rationality, incentive compatibility for both winning travelers and TSPs. Meanwhile, it guarantees weakly budget balance such that the IRSP will not subsidize the auction. We present the corresponding propositions as below and the detailed proofs can be found in the supplementary material (refer to Appendix 1-3).

**Proposition 1.** *The proposed online mechanism M1 guarantees individual rationality (IR) and incentive compatibility (IC) for each traveler. Specifically, each traveler will obtain non-negative utility from the auction and bidding truthfully is the dominant strategy.*

**Proposition 2.** *The proposed online mechanism M1 guarantees individual rationality (IR) and incentive compatibility (IC) for each TSP. Specifically, each TSP will obtain non-negative utility from the auction and bidding truthfully is the dominant strategy.*

**Proposition 3.** *The proposed online mechanism M1 guarantees weakly budget balance. Specifically, the total payments collected from travelers can cover the total expenditures distributed to TSPs.*

To facilitate the understanding of the above online mechanism, we present an illustrative example (Fig. 3) to demonstrate the process. Consider a certain OD pair; Tables 1 and 2 list the bid information from travelers and TSPs. We assume that travelers can be categorized into different groups and that members of the same group have identical bidding information; for example, travelers 1 and 6 are in the same group. For each TSP, we assume that it would bid the same operational cost for a specific service type within a short time span; for example, TSP B's bids 3 and 5 have the same declared operational cost.

In the first time slot $\mathbb{t}_1$, three travelers (1, 2 and 3) and two TSPs (A and B) submit bids. With these bids, we solve the social welfare maximization problem defined in Section 3 to determine the critical prices. In the optimal solution, traveler 1 is matched to bid 1 with real value of \$21 as calculated by Eq. (1). Traveler 2 and 3 are matched to bid 4 with real value of \$18. Then the critical prices are \$21 for the luxury service and \$18 for the economic service. Meanwhile the highest operational costs for the luxury and economic services are \$15 and \$10, respectively.

In the second time slot $\mathbb{t}_2$, three travelers (4, 5 and 6) and three TSPs (A, B and C) submit bids. We need to determine the matching and pricing with the critical prices obtained in $\mathbb{t}_1$. First, the potential capacity $\overline{\boldsymbol{W}}_{\mathbb{t}_2}$ in this time slot is computed. Bid 6 has the chance to win the bid (lower declared operational cost than \$15), while bid 5 and bid 7 do not. Therefore, we have $\overline{\boldsymbol{W}}_{\mathbb{t}_2} = [1,0]$. If bid 6 wins, the trading price will be \$15. Travelers will be considered one by one according to the arriving order. Traveler 4 gets greater real value with the luxury service (\$22) than the economic service (\$19). Then the maximal achievable real value of traveler 4 is \$22, which is greater than the critical price \$21. So, traveler 4 will be accepted and served by bid 6′s vehicle with payment of \$21 (the greater amount between the critical price of \$21 and the unit trading price of TSP \$15). After accepting traveler 4, the potential capacity is updated to $\overline{\boldsymbol{W}}_{\mathbb{t}_2} = [0,0]$, indicating that no service is available. Then, both travelers 5 and 6 are rejected. At the end of $\mathbb{t}_2$, the available capacity that will be passed to the subsequent time slot $\mathbb{t}_3$ to determine $\overline{\boldsymbol{W}}_{\mathbb{t}_3}$ remains $\boldsymbol{W} = [0,0]$.

### 4.2. Online mechanism designed with a greedy algorithm

For comparison purpose, in this section we introduce another online mechanism based on the greedy algorithm. Greedy algorithm is a popular solution method to online problems. The basic idea is simple: in each time slot, sort the bids from travelers and TSPs according to a certain order, conduct matching to maximize the social welfare and determine the trading price according to rejected agents in the matching. Fig. 4 demonstrates the procedure of our proposed mechanism with a greedy algorithm, followed by the details in each step.

- Step 1: **Initialization**. Let $\mathbb{T}$ denote the total time span, which is divided to $\mathcal{M}$ time slots, denoted as $\mathbb{t}_1, \mathbb{t}_2, \ldots, \mathbb{t}_m$. The length of each time slot is identical, which can be regarded as the pooling time to collect current TSPs' and travelers' bids.

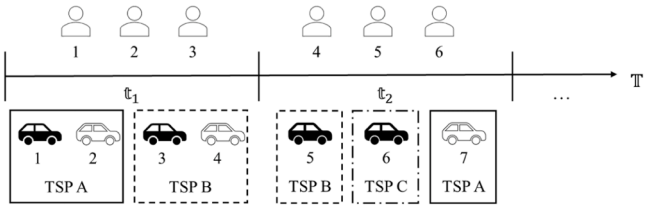



**Fig. 3.** Illustration of the online requests.

**Table 1**
Travelers' bidding information.

| Traveler | $b_i$(\$) | $T_i$ | $R_i$ | $\gamma_i$ | $\beta_i$ |
|---|---|---|---|---|---|
| 1,6 | 25 | 10 | 1 | 0.8 | 0.8 |
| 2,3,5 | 18 | 20 | 3 | 0.2 | 0.3 |
| 4 | 22 | 15 | 1 | 0.5 | 0.5 |

**Table 2**
TSPs' bidding information.

| Bid | TSP | Service type | $t_k$ | $r_k$ | $C_k$(\$) |
|---|---|---|---|---|---|
| 1 | A | | | | 15 |
| 3,5 | B | Luxury | 15 | 1 | 18 |
| 6 | C | | | | 13 |
| 2,7 | A | | | | 15 |
| 4 | B | Economic | 20 | 2 | 10 |

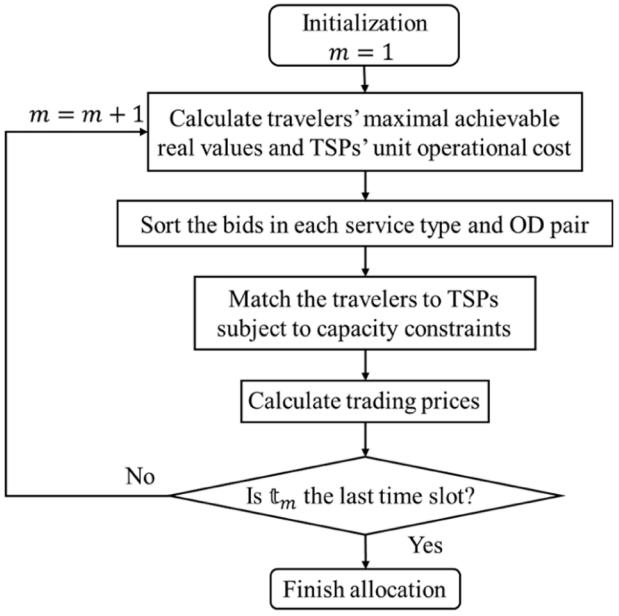



**Fig. 4.** Online mechanism with a greedy algorithm.

- Step 2: **Sorting.** Collect all the bids from agents arrived by $\mathbb{t}_1$. Calculate each traveler's maximal achievable real value given the available service types provided by the TSPs in $\mathbb{t}_1$. The corresponding service type would be assigned to the traveler if he or she ultimately wins the bid. Calculate the unit operational cost for each bid submitted by TSPs, which is the operational cost divided by the rider capacity. For each service type of an OD pair, sort travelers' bids from high to low based on their maximal achievable real values and sort TSPs' bids from low to high based on unit operational cost.
- Step 3: **Matching.** Conduct matching between travelers and TSPs according to the order subject to capacity constraints. Once the capacity of a certain service type of a certain OD pair is used up, return to Step 2 to update the maximal achievable real values of the remaining travelers and sort again.
- Step 4: **Trading prices calculation.** Traveler's payment for each OD pair and service type corresponds to the highest real value among all rejected travelers of that OD pair and service type. If there are no rejected travelers, the accepted traveler with the lowest real value will be rejected, and his or her real value will be used as the payment for the other travelers. This traveler will also be rejected if the payment is less than the unit operational cost of the matched transport service. Rejected travelers are removed from the auction and are not

allowed to rebid. The trading price for each bid from TSPs is equal to the smaller one between the total payments of travelers matched to the vehicle and the lowest reported operational cost among all the rejected bids from other TSPs (excluding itself) with the same OD pair and service type. Rejected bids are removed from the auction, and TSPs are not allowed to rebid on rejected vehicles.

- Step 5: **Iteration.** Repeat Steps 2–4 for the remaining time slots $\mathbb{t}_2$, $\mathbb{t}_3$, …, $\mathbb{t}_m$.

It should be noted that the above greedy-based online mechanism does not truly realize immediate response to travelers. It sorts all the bids from travelers and TSPs that arrive in one time slot, then decides the allocation and pricing. As a result, travelers and TSPs have to wait for a while for the decision to come out. The length of time slot greatly influences the performance of the mechanism. Intuitively, longer time slot achieves more effective allocation with higher social welfare. However, longer time slot also indicates longer waiting time experienced by travelers and TSPs, which will harm their satisfaction to the service and cause user defection to IRSP in the long run.

Let M2 denote the online double auction mechanism designed with a greedy algorithm. We claim that M2 achieves individual rationality, incentive compatibility for both winning travelers and TSPs. M2 also guarantees weakly budget balance. We present the corresponding proposition as below and the detailed proofs can be found in the supplementary material (refer to Appendix 4-6).

**Proposition 4**. *The proposed online mechanism M2 guarantees individual rationality (IR) and incentive compatibility (IC) for each traveler. Specifically, each traveler will obtain non-negative utility from the auction and bidding truthfully is the dominant strategy.*

**Proposition 5**. *The proposed online mechanism M2 guarantees individual rationality (IR) and incentive compatibility (IC) for each TSP. Specifically, each TSP will obtain non-negative utility from the auction and bidding truthfully is the dominant strategy.*

**Proposition 6**. *The proposed online mechanism M2 guarantees weakly budget balance. Specifically, the total payments collected from travelers can cover the total expenditures distributed to TSPs.*

We continue to use the example from Section 4.1 to clarify the procedure of the online mechanism with a greedy algorithm. In this mechanism, the matching in each time slot is independent. In the first time slot $\mathbb{t}_1$, we start with sorting the bids. According to their maximal achievable real value, the rank of travelers should be: traveler 1 ($21) > traveler 2 ($18) = traveler 3 ($18). According to TSPs' unit operational costs, bid 1 ($15) should rank above bid 2 ($18) for the luxury service, while bid 4 ($5) should rank above bid 2 ($7.5) for the economic service. Then, traveler 1 will be matched to bid 1, and travelers 2 and 3 will both be matched with bid 4. However, when calculating the trading prices, since no traveler is rejected, these three travelers will be rejected according to Step 4 of M2. This indicates that the greedy mechanism performs poorly when supply exceeds demand.

For comparison purpose, we further propose an online mechanism based on the framework of the greedy mechanism. This approach is developed with a padding method, in which for each time slot, we solve an offline problem with the existence of virtual padding travelers. It has been proved that this padding method can achieve asymptotic social welfare maximization when supply and demand are sufficiently large (Chu, 2009). We compare the critical price dynamic updating mechanism and the greedy mechanism with this padding mechanism in numerical studies. The detailed description of this mechanism and the corresponding proofs of propositions can be found in Appendix 9–12 of the supplementary material.

## 5. Numerical experiments

In this section, we evaluate the performance of our proposed online mechanisms by comparing to the first-arrive-first-serve (FAFS) matching. FAFS algorithm simply matches the travelers and bids from TSPs according to the arrival order. Travelers' and TSPs' bids are accepted if the travelers' real values are greater than or equal to the operational cost of the vehicle. Note that theoretical proving the bounds of efficiency loss is not feasible since we have no assumption on the arrival order of agents, e.g., random permutation Ding et al., 2023). Then the efficiency loss of the worst-case scenario is actually unbounded. Hence, in the following analysis, we first evaluate the performance of three online mechanisms to the FAFS algorithm under various supply-demand scenarios, which reveals inherent trade-offs between the number of agents and the resultant efficiency loss. Then, we compare the individual TSPs' revenues with and without an IRSP to demonstrate the benefits of the IRSP's integration of mobility resources. Further, we examine two market structure scenarios to identify the effects on individual TSPs: ((1) a large TSP and a small TSP; (2) two TSPs of comparable size.

### 5.1. Experimental setup

We conduct the numerical test on a network with four OD pairs. Ten TSPs operate in the transport service market with two service types, i.e., luxury and economic, on each OD pair. The luxury service has shorter travel time and fewer riders sharing one vehicle than the economic service. We generate the bidding prices of travelers following a log-normal distribution, with the mean 3.7 and variance 0.3. Then, we sort the values and assign them to travelers based on their preferred travel time and the number of shared riders. In general, a traveler's bid will be higher if he or she requests a faster transport service with fewer passengers. There are, however, a few exceptions to this rule, such as the fact that some travelers are less sensitive to price and are willing to pay more even if their requested travel time is long. Hence, we randomly modify the bidding prices of several travelers to make exceptions. For TSPs, their bidding operational costs are estimated from historical operation data. OD pair with long travel distance naturally would have higher operational cost. The total operating time span for the online case is 20 min.

Two metrics are adopted to evaluate the performance of online mechanisms: ratio of social welfare between the online mechanism and the optimal allocation (*RSW*), ratio of matching rate between the online mechanism and the optimal allocation (*RMR*). The optimal allocation is obtained by solving $P(I, K)$ in Section 3 with all the bids from all agents in the operating time span. Note that such allocation is not implementable because there is no corresponding real-time pricing scheme ensuring the desired economic properties. Thus, it only provides an upper bound of the achievable social welfare and serves as the benchmark. The matching rate denotes the percentage of accepted travelers in all bidding travelers. We conduct the experiments on a laptop of Windows 10, Intel(R) Core i5–6200 U CPU @ 2.30 GHz, 4 Core(s) with RAM of 8 GB.

### 5.2. Experimental results

First, we examined six supply-demand scenarios: 500, 1000 and 1500 travelers, with a total capacity of 1000 on the TSP side; 1000, 2000 and 3000 travelers, with a total capacity of 2000 on the TSP side. Note that these are all the bids during the operating time span, including all the OD pairs. The time slot for the critical price dynamic updating mechanism and the greedy mechanism is set to two minutes. (Note: the length of the time slot represents pooling time for the greedy mechanism, and the sensitivity analysis result on the pooling time is presented in Appendix 13 of the supplementary material.) Twenty instances were generated with varying bid arrival orders from agents. Table 3 and Table 4 present the mean *RSW* and *RMR* of three online mechanisms

**Table 3**
Comparison of online mechanisms (10 TSPs, Total capacity=1000).

| | Demand of travelers | Padding method | Greedy | Critical price dynamic updating | First arrive first serve |
|---|---|---|---|---|---|
| *RSW* | 500 | **0.839** | 0.550 | 0.716 | 0.347 |
| | 1000 | 0.686 | 0.554 | **0.717** | 0.283 |
| | 1500 | 0.655 | 0.732 | **0.789** | 0.250 |
| *RMR* | 500 | **0.806** | 0.689 | 0.590 | 0.486 |
| | 1000 | 0.599 | **0.648** | 0.570 | 0.377 |
| | 1500 | 0.522 | **0.766** | 0.632 | 0.361 |
| Computation time | 500 | 2043.50s | 58.58s | 105.10s | 13.92s |
| | 1000 | 2365.33s | 120.95s | 224.50s | 33.72s |
| | 1500 | 2899.75s | 167.72s | 244.27s | 47.64s |

**Table 4**
Comparison of online mechanisms (10 TSPs, Total capacity=2000).

| | Demand of travelers | Padding method | Greedy | Critical price dynamic updating | First arrive first serve |
|---|---|---|---|---|---|
| *RSW* | 1000 | **0.917** | 0.572 | 0.741 | 0.244 |
| | 2000 | **0.864** | 0.587 | 0.776 | 0.346 |
| | 3000 | 0.726 | 0.654 | **0.806** | 0.339 |
| *RMR* | 1000 | **0.882** | 0.743 | 0.619 | 0.341 |
| | 2000 | **0.766** | 0.621 | 0.620 | 0.466 |
| | 3000 | 0.579 | **0.682** | 0.609 | 0.334 |
| Computation time | 1000 | 3586.22s | 167.27s | 368.77s | 27.67s |
| | 2000 | 3882.36s | 355.08s | 490.99s | 39.92s |
| | 3000 | 4204.60s | 600.18s | 711.47s | 50.03s |

with these 20 instances.

As shown in Table 3, the padding mechanism achieves the highest ratio of social welfare *RSW* in the first demand scenario, i.e., when the demand is less than the supply. However, the computation time of the padding mechanism significantly exceeds the online operation time, which hinders its applicability to the real-world matching and pricing for the IRSP. Hence, only the critical price dynamic updating mechanism and greedy mechanism are considered implementable in practice. In the following discussion, we mainly focus on these two mechanisms. As the demand rises, the performance of the critical price dynamic updating mechanism remains stable, the greedy mechanism improves steadily, while the padding mechanism performs worse. For the greedy mechanism, this is due to the diminishing negative impact of pre-dividing time slots. When the demand is less than the supply, the greedy algorithm may experience an imbalance of supply between different pre-divided time slots, e.g., time slot 1 may be oversupplied while time slot 2 is in short supply; however, according to the greedy mechanism, the excess supply in time slot 1 cannot satisfy the demand in time slot 2. This causes efficiency loss. As the demand increases, we can consider each time slot to be in supply shortage, and the impact of pre-dividing time slots becomes negligible. For the padding mechanism, due to the fact that the introduction of virtual padding travelers intensifies the competition among travelers, those who could have been served if not for the virtual padding travelers are rejected to guarantee incentive compatibility. When the demand is greater than the supply, such deliberate rejection results in diminished social welfare more significantly.

Although *RSW* of the greedy mechanism and the critical price mechanism are less than 80 %, which seems not ideal, we evaluate their performance by comparing it to the FAFS mechanism because *RSW* is obtained by dividing the results with the optimal matching solved from the static problem with all the bids known as input, which is impossible to be realized in the online case. In all three supply-demand scenarios, both mechanisms improve *RSW* significantly, by an average of 159.4 % for the critical price dynamic updating mechanism and 116.7 % for the greedy mechanism. The greedy algorithm achieves the highest ratio of matching rate *RMR* in most demand scenarios because it utilizes all the available resources in each time slot without taking price constraints into account during the allocation process. Both online mechanisms perform significantly better than FAFS, and the improvement is more pronounced when demand exceeds supply.

Note that the computation time in Table 3 is the total amount of time required to process bids from travelers and TSPs in 20 min for all three algorithms. Both online mechanisms run in polynomial time, and the corresponding analysis of time complexity is provided in the supplementary material (refer to Appendix 7-8). It should be noted that the waiting time for individual traveler and TSP does not equal the computation time. In the critical price dynamic updating mechanism, for instance, travelers can receive a response in real time without having to wait because the process of updating the critical price is conducted at the end of the previous time slot. Upon receiving a traveler's request, the platform is able to make instantaneous allocation and pricing decisions based on current critical prices and capacity. In contrast, the greedy mechanism and the padding mechanism require travelers to wait the pooling time (in this case, two minutes) because the matching is conducted after all bids are collected in the current time slot. In FAFS, travelers and TSPs receive real-time responses, but this matching rule does not guarantee incentive compatibility or individual rationality. The social welfare obtained by FAFS can be considered the results without optimization and is used solely for comparison with the two online mechanisms we propose.

Then we evaluate three mechanisms' performance in larger supply-demand scenarios, i.e., 1000, 2000 and 3000 travelers, with a total capacity of 2000 on the TSP side. The results are summarized in Table 4. Likewise, the padding mechanism outperforms the other two when the demand is no larger than the supply, yet the computation time is too long to be acceptable for online mechanisms. Comparing the critical price dynamic updating mechanism and the greedy mechanism, the former always achieves higher *RSW* while the latter always achieves higher *RMR*. Comparing the results of Table 3 and Table 4, it is concluded that the critical price dynamic updating mechanism performs better with larger demand and supply, which is due to the fact that with more bids from travelers and TSPs, the critical price will be more accurate, leading the matching closer to the optimum.

Next, we will discuss the advantages of introducing an IRSP. Without an IRSP, travelers first decide which TSP to request service from, and we assume they only submit bids to a single TSP, which is consistent with reality. To demonstrate the impacts of introducing the IRSP to individual TSPs clearly, we assume only two TSPs operating the transport network and investigate two market structure scenarios: (1) a large TSP and a small TSP; (2) two TSPs of comparable size. Under the first market structure, we assume that 70 % of travelers choose the large TSP and 30 % choose the small TSP. This is consistent with the fact that travelers typically prefer the services of large TSPs over those of small ones. For the second market structure (two TSPs of comparable size), we set the proportions to 0.5 and 0.5, indicating that the probability of travelers choosing one TSP is identical to that of the other. We employ the critical price dynamic updating mechanism to solve the online matching and compare the revenues of two TSPs with and without an IRSP. Three demand scenarios are tested and the results are summarized in Table 5.

For the first market structure, the revenue of the large TSP A increases in all three demand scenarios, whereas the revenue of TSP B decreases when supply exceeds demand (500 travelers case). This is because in the oversupply case, the service provided by the small TSP with higher operational costs is inferior to that provided by the large TSP with lower operational costs to the IRSP. The integrated platform prefers to accept bids from the large TSP over those from the small TSP. However, if there is no IRSP, the small TSP will have a fixed proportion of demand and will be able to serve these travelers despite having higher operational costs. Thus, in the oversupply case, the revenue of the small TSP B decreases after the introduction of IRSP. Under this market structure, the introduction of IRSP benefits the large TSP more than the

**Table 5**
The revenues of two TSPs with and without an IRSP.

| | Demand scenario | | 500 | 1000 | 1500 |
|---|---|---|---|---|---|
| The first market structure | Without an IRSP | TSP A (big) | 588.16 | 2004.48 | 3041.68 |
| | | TSP B (small) | 238.00 | 694.02 | 950.26 |
| | With an IRSP | TSP A (big) | 3333.16 | 5272.76 | 5651.38 |
| | | TSP B (small) | 51.96 | 1025.25 | 1839.53 |
| The second market structure | Without an IRSP | TSP A | 255.20 | 511.00 | 753.20 |
| | | TSP B | 214.20 | 508.00 | 695.40 |
| | With an IRSP | TSP A | 1624.24 | 3454.08 | 4408.13 |
| | | TSP B | 1726.41 | 3553.44 | 4397.88 |

small TSP.

In the second market structure, both TSPs receive higher revenues with an IRSP in each of the three demand scenarios, and their improvements are comparable. Intriguingly, the IRSP increases the revenues of two TSPs more than in the first market structure, indicating that the IRSP in the second market structure is better able to integrate mobility resources to balance supply and demand. For instance, if one TSP experiences a supply shortage while the other has excess resources, and both TSPs operate at a similar level, their services are substitutes, and when pooled by the IRSP, the imbalance between supply and demand faced by the two TSPs is eliminated. In the first market structure, however, the services of two TSPs are not exact substitutes because their operational costs are different. Thus, even when pooling the supply and demand together, the bids from the small TSP with higher operational costs may still be rejected, resulting in a revenue increase that is less than that of the second market structure.

## 6. Conclusion

In this paper, we present a double auction-based framework to solve the matching and pricing problems for an IRSP that provides on-demand transport services. To provide personalized services to travelers, the mechanisms we propose enable travelers to report their multi-dimensional travel needs (bidding price, preferred travel time, preferred number of shared riders). To make the problem more realistic, TSPs are allowed to submit bids on multiple OD pairs with multiple service types. Such a setting transforms our problem into a more complex double auction mechanism design with multi-minded buyers and sellers with multi-units. In addition, since the IRSP operates in an online environment in practice, matching and pricing decisions should be made in real time.

We develop three mechanisms, one based on a critical price dynamic updating algorithm, one on a greedy algorithm and one based on a padding method, all of which are theoretically proved to be truthful, individually rational, and weakly budget balanced. To assess the performance of the proposed mechanisms, we conduct a numerical analysis and compare them to the FAFS matching rule in various supply-demand scenarios. The results indicate that all of our proposed online mechanisms improve social welfare substantially compared to FAFS in all demand scenarios. Additionally, we examine the effect of introducing an IRSP on the revenues of individual TSPs under two market structures. For the asymmetric market consisting of a large TSP and a small TSP, the IRSP provides greater benefits to the large TSP than the small TSP. The small TSP may even experience a decline in revenue in the event of oversupply. For the symmetric market, which consists of two TSPs of comparable size, the IRSP increases the revenues of both TSPs to a greater extent than in the asymmetric market.

There are multiple avenues worthy of further investigation. First, in this study, we only consider the integration of ride-sourcing companies, whose services are OD-based. The computational burden has already been heavy for large-scale matching. However, when public transit services are incorporated into the integrated platform, this problem becomes link-based. The link-path generation in a transport network should be discussed, given that multimodal services are less flexible than ride-sourcing, which further increases computation time. More efficient algorithm is required to achieve real time responses. Second, we assume that travelers and TSPs will not cheat on their bidding time. This assumption is plausible, but there are real-world exceptions: non-myopic agents might intentionally delay their bids to obtain higher utilities. Besides, we assume TSPs cannot negotiate the market price through strategic collaboration because we deem profitable direction difficult to capture. Nevertheless, some smart agents may achieve it and benefit from manipulation. Relaxation of these assumptions requires additional design of pricing strategies to guarantee incentive compatibility. Third, from the practical standpoint, successful matching is not the end of the service, but the beginning. The matching of TSPs and travelers is performed by drivers. It is worthwhile to investigate the possibility of minimizing travelers' waiting time through efficient routing in a transport network, particularly when a vehicle is allowed to serve multiple passengers with different origins. In addition, the impact of drivers' multi-homing behavior, i.e., serving multiple TSPs simultaneously, can be another avenue for study. From travelers' perspectives, a wider variety of personalized preferences, e.g., flexibility needs, can be incorporated in future studies, as long as they can be quantified in the valuation functions. Finally, the lack of theoretical proof of efficiency loss for online mechanisms is a limitation of this study which warrants further investigation.

## CRediT authorship contribution statement

**Xiaoshu Ding:** Investigation, Methodology, Writing – original draft, Writing – review & editing. **Qi Qi:** Conceptualization, Investigation, Methodology, Supervision, Writing – review & editing. **Sisi Jian:** Conceptualization, Investigation, Methodology, Supervision, Funding acquisition, Writing – original draft, Writing – review & editing.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ejor.2023.12.004.

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